

Uncertainty-Conditioned Reinforcement Learning with Evidential Policy Networks for Autonomous Driving

Antonio Galdes

Abstract—While recent work has separately applied uncertainty quantification methods to perception systems and used reinforcement learning for control, the integration of explicit localisation uncertainty into learned control policies that produce calibrated uncertainty outputs has received limited attention. This study suggests an architecture that combines two methods for handling uncertainty: using evidential deep learning in the policy network to produce calibrated epistemic and aleatoric uncertainty estimates, and supplying Simultaneous Localisation and Mapping (SLAM) covariance matrices as explicit state input to reinforcement learning policies. By enabling the control strategy to modify its behaviour in response to both internal model uncertainty and external localisation confidence, the suggested architecture is intended to facilitate safer navigation. This approach will be validated by using systematic simulation tests to show how well it handles localisation degradation.

Index Terms—Evidential Deep Learning, Reinforcement Learning, Autonomous Driving, Uncertainty Quantification, SLAM

I. INTRODUCTION

Three primary components are used by autonomous driving systems: control to carry out driving operations, planning to identify the best routes, and perception to interpret the surroundings [1]. However, a major problem emerges when intrinsic uncertainties from sensor noise (aleatoric uncertainty) and inadequate information during training (epistemic uncertainty) spread across this pipeline [2]. For safe deployment, it is crucial that control policies react to these uncertainties correctly.

Numerous quantification techniques are used across system components to overcome these issues. Consequently, current Reinforcement Learning (RL) techniques rarely integrate calibrated uncertainty estimates in policy outputs with explicit localisation uncertainty as policy input. By leveraging the uncertainty that SLAM systems naturally quantify through covariance matrices, control policies can adapt their behaviour based on localisation confidence. Moreover, these policies can produce outputs that reflect their own uncertainty.

By learning the parameters of higher order probability distributions, Evidential Deep Learning (EDL) has become a popular method for quantifying uncertainty [3]. Evidential networks compute both aleatoric and epistemic uncertainty in a single forward pass, in contrast to sampling based techniques like Monte Carlo Dropout or Deep Ensembles that necessitate several computationally costly inference steps. This real time efficiency is particularly valuable for control policies operating

at high frequencies, as it enables the system to adopt cautious behaviours or revert to human operation.

Despite these clear advantages for real time decision making, evidential methods have primarily been demonstrated in perception tasks such as depth estimation and 3D object detection for autonomous vehicles [4]. However, their application to learned control policies in the autonomous driving pipeline remains largely unexplored.

II. AIM AND OBJECTIVES

Motivated by the objective of enhancing safety in autonomous navigation, this research is situated at the intersection of uncertainty aware control and evidential deep learning. In addition, most current research either applies evidential approaches to critic networks instead of policy actors or conditions policies on localisation uncertainty without producing evidential outputs. Thus, combining evidential neural network policy outputs with SLAM uncertainty as an explicit RL state input offers a new research direction that is still mainly unexplored.

Consequently, this research addresses the following primary research question:

How can autonomous vehicle control policies explicitly incorporate and respond to localisation uncertainty from SLAM systems, and what benefits arise from producing calibrated uncertainty estimates through evidential policy outputs?

A. Aim

The main aim of this proposed project is to develop and validate an uncertainty conditioned reinforcement learning architecture that uses EDL in the policy network to produce calibrated uncertainty estimates for autonomous driving control, demonstrating improved safety metrics when localisation quality degrades.

B. Objectives

To achieve the primary aim, the following objectives have been established:

- **Localisation & Uncertainty Extraction:** Implement an EKF-based pipeline in CARLA to provide covariance-derived uncertainty features for the RL state.
- **Evidential RL Development:** Develop an Evidential PPO algorithm using NIG distributions for single-pass uncertainty quantification in steering and throttle control.

- **Comparative Performance Analysis:** Evaluate the system against standard PPO and uncertainty-conditioned baselines, focusing on safety and calibration under degraded sensor conditions.

III. BACKGROUND

A. Technologies

The CARLA simulator [5] provides an open source urban driving environment with realistic physics, sensor simulation, and configurable weather conditions. In order to accomplish Extended Kalman Filter (EKF) sensor fusion and publish filtered odometry with pose covariance matrices, this study will make use of CARLA's integration with Robot Operating System (ROS) 2 Jazzy. The neural network input then uses these as explicit uncertainty features.

B. Techniques

1) *Evidential Deep Learning:* Standard neural networks for control typically output point estimates or simple distributions that conflate data noise with model ignorance. By applying a higher order prior over the output distribution parameters, EDL overcomes this restriction and permits the separation of aleatoric and epistemic uncertainty [6].

For continuous control, the network predicts the parameters of a Normal Inverse Gamma (NIG) distribution $(\gamma, \nu, \alpha, \beta)$, in which γ denotes the predicted control action. The remaining parameters encode predictive uncertainty, which can be decomposed into aleatoric and epistemic components:

$$\sigma_{\text{aleatoric}}^2 = \frac{\beta}{\alpha - 1}, \quad \sigma_{\text{epistemic}}^2 = \frac{\beta}{\nu(\alpha - 1)}. \quad (1)$$

Aleatoric uncertainty captures irreducible observation noise. In contrast, epistemic uncertainty reflects model uncertainty arising from limited or unfamiliar training data [3].

2) *Proximal Policy Optimisation:* Proximal Policy Optimisation (PPO) is a policy gradient method that improves training stability by constraining policy updates through a clipped surrogate objective [7]. Its balance between stability and sample efficiency makes it well suited for continuous control tasks.

In evidential control settings, PPO provides a flexible optimisation framework for policies that output distributional action representations. During training, the actor objective may be augmented with an evidential negative log likelihood term. Additionally, regularisation components discourage overconfident predictions through preserving the core PPO structure.

3) *EKF Based State Estimation:* The EKF mitigates sensor noise and drift by fusing measurements from multiple sources into a unified state estimate [8]. These sources may include Global Positioning System (GPS), lidar, wheel odometry, and Inertial Measurement Units (IMUs). Alongside the estimated pose, the EKF maintains a covariance matrix that provides a quantitative measure of localisation uncertainty. Recent work has improved learned covariance estimation for visual odometry [9], [10], demonstrating that such uncertainty estimates can benefit decision making in autonomous systems. However,

explicit integration of these naturally available covariance estimates into learned control policies remains unexplored.

C. Related Work

1) *GUIDE/G SAC:* The work in [11] presents Task-Specific Uncertainty Maps (TSUMs), which is produced by Contrastive Language-Image Pre-training (CLIP) based vision language processing and defines uncertainty levels across spatial regions based on task requirements. These task derived thresholds and the robot's present state estimation uncertainty are incorporated into their Soft Actor Critic (SAC) framework. Although this illustrates the usefulness of uncertainty conditioned policies, the suggested method of directly utilising SLAM derived covariance matrices contrasts with their reliance on computationally costly vision language models for uncertainty map construction. Without the need for task specific retraining, using uncertainty estimates that naturally arise as byproducts of sensor fusion may provide benefits in terms of computational efficiency and generalisability across tasks. Furthermore, the deconstructed epistemic and aleatoric uncertainty estimations that evidential networks offer are absent from their use of SAC outputs.

2) *Active SLAM with Deep RL:* The approach in [12] formulates active SLAM by embedding the D optimality criterion of SLAM covariance matrices into the reward function. Uncertainty estimations from gmapping's particle distribution are incorporated into the duelling double Deep Q-Network (DQN) architecture as an intrinsic reward signal proportionate to the covariance matrix determinant. This demonstrates the value of SLAM uncertainty metrics in RL. However, the approach differs fundamentally in objective and mechanism. The system learns to actively seek locations that reduce uncertainty through exploration and loop closure. In contrast, the proposed work conditions policy inputs on measured uncertainty to adapt behaviour conservatively when localisation quality degrades.

3) *EPPO:* The method in [13], which represents the state of the art in evidential reinforcement learning, employs Normal Inverse Gamma distributions in the *critic* to preserve plasticity and facilitate directed exploration in non stationary settings. Nevertheless, this method limits evidential uncertainty to the value function rather than the policy actor itself. Moreover, the study ignores the upstream localisation or SLAM uncertainty characteristic of autonomous driving, focussing instead on standard continuous control benchmarks such as MuJoCo.

4) *CEQR DQN:* Similarly, the work in [14] combines evidential deep learning with distributional RL for Q function estimation, successfully separating aleatoric and epistemic uncertainty to improve the exploration exploitation balance. Like EPPO, this method applies evidential learning to value estimation through Q values rather than policy action distributions. Additionally, the evaluation is limited to discrete action spaces in Atari games, diverging from the continuous control and localisation challenges inherent to this project.

5) *DRO EDL MPC:* In the domain of autonomous driving, the study in [15] demonstrates the utility of EDL by using ev-

identical perception uncertainty to construct ambiguity sets for distributionally robust control. However, this system utilises Model Predictive Control (MPC) rather than reinforcement learning. Furthermore, the evidential components quantify perception uncertainty through the world model rather than the uncertainty of a learned policy.

6) *Evidential Reward Estimation*: Finally, the work in [16] explores evidential networks as a method for quantifying uncertainty in RL reward estimation. This highlights the versatility of evidential methods. Nevertheless, the application is distinct from policy learning and does not address the specific challenges of localisation conditioned control in robotics.

7) *Ensemble Methods for Uncertainty Quantification*: Recent safe RL approaches utilise deep ensembles for epistemic uncertainty estimation to enable adaptive safety constraints during autonomous driving [17]. These methods demonstrate the importance of uncertainty aware control. However, ensemble based approaches require multiple forward passes during inference, which may limit real time control applications. Additionally, these approaches do not integrate upstream localisation uncertainty from SLAM systems. The proposed work explores whether combining EKF covariance inputs with evidential policy networks could enable efficient single pass uncertainty quantification for adaptive control under varying localisation quality.

IV. PROPOSED METHODOLOGY

The proposed architecture, illustrated in Figure 1, integrates localisation uncertainty into an uncertainty aware RL control framework.

The pipeline starts with environmental data generated by onboard vehicle sensors, which the SLAM system processes to provide a map, an estimated vehicle pose, and associated localisation uncertainty produced from EKF covariance matrices.

Using the current map and pose, the Goal Manager establishes intermediate targets in the direction of the user specified

final destination. Subsequently, the Path Planner computes collision free reference trajectories to these intermediate goals. This allows the vehicle to adapt as new environmental information becomes available.

These reference trajectories, together with the estimated vehicle state and localisation uncertainty, are provided as inputs to an evidential RL policy. Rather than producing deterministic control commands, the policy outputs actions parameterised by a predictive distribution. Consequently, uncertainty estimates are derived from this distribution, enabling downstream components to reason explicitly about both confidence and risk.

Before sending the suggested control commands to the vehicle actuators during deployment, a Safety Monitor assesses them together with the uncertainty estimations that accompany them. This distinction allows for tailored interventions. For instance, high aleatoric uncertainty reflecting inherent environmental noise may trigger conservative adjustments like speed reduction. In contrast, high epistemic uncertainty indicating model ignorance in novel scenarios could necessitate requesting human takeover.

A. Challenges and Limitations

1) *CARLA ROS Integration*: Synchronising CARLA simulation time with ROS 2 message timestamps requires some configuration. The CARLA ROS bridge may introduce latency that affects EKF performance.

2) *Evidential Training Stability*: Evidential loss functions can exhibit instability during early training when evidence values are poorly initialised. Therefore, regularisation strategies from the literature will be employed to mitigate this issue.

3) *Covariance Feature Scaling*: EKF covariance values span multiple orders of magnitude depending on sensor quality and filter convergence. Appropriate normalisation strategies must be developed for effective neural network training.

4) *Reward Function Design*: Designing a reward function that appropriately balances task completion, safety, and uncertainty aware behaviour presents a major challenge. The reward

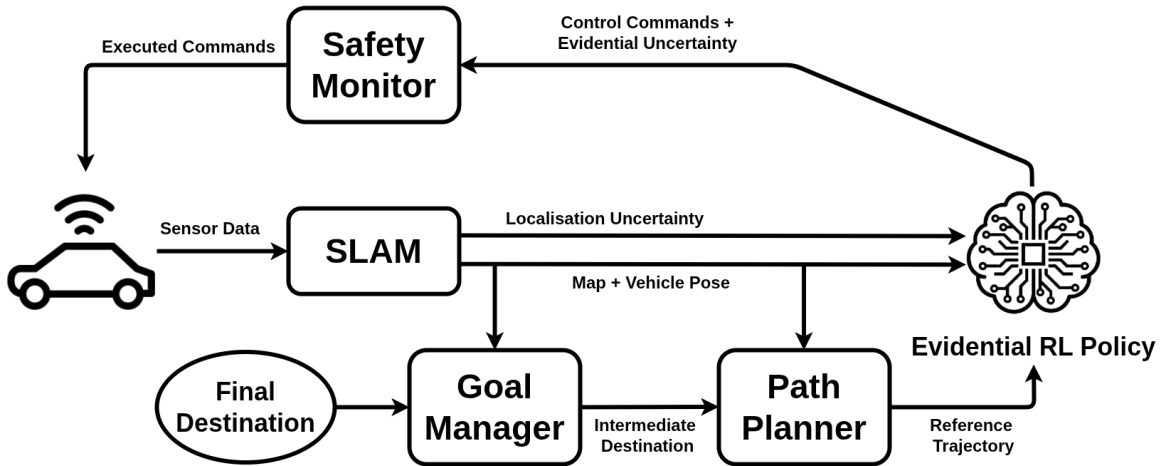


Fig. 1: System architecture showing the uncertainty conditioned control pipeline.

should motivate the policy to both accomplish navigation goals and react cautiously to high localisation uncertainty. However, this must be achieved without unnecessarily penalising advancement. Additionally, the balance between these conflicting goals needs to be carefully adjusted and may change depending on the driving situation.

5) *Pipeline Dependency and Latency*: Since the Evidential RL policy requires the reference trajectory from the Path Planner as a prerequisite input, a sequential dependency is introduced. The policy must wait for the planning cycle to complete before computing control actions. This cascading latency could propagate through the pipeline and affect control stability. Therefore, efficient synchronisation will be necessary to mitigate this risk.

V. TESTING AND EVALUATION

A. Evaluation of Objective 1: Localisation Pipeline

The EKF based localisation pipeline will be evaluated by comparing estimated vehicle poses against ground truth trajectories provided by CARLA. Root mean square error in position and orientation will be computed under nominal sensor conditions to establish baseline performance. Subsequently, systematic sensor degradation will be introduced through increasing noise levels in GPS, IMU, and odometry measurements. The quality of covariance estimates will be assessed by examining whether they accurately reflect true localisation error magnitudes [18]. This ensures that uncertainty features provided to the policy network are reliable indicators of localisation confidence.

B. Evaluation of Objective 2: Evidential Policy Development

The evidential PPO algorithm will be evaluated on trajectory following tasks in controlled driving scenarios. These may include parking manoeuvres and navigation tasks appropriate for the simulation environment. Lateral path tracking error will measure deviation from reference trajectory centrelines, in addition to longitudinal tracking error for speed profile adherence. The calibration quality of evidential uncertainty estimates will be evaluated using Expected Calibration Error (ECE), which measures the alignment between predicted confidence and actual accuracy [19]. Furthermore, additional metrics may include sharpness of uncertainty estimates and their correlation with observed tracking errors. Consequently, these evaluations will determine whether the policy produces well calibrated uncertainty outputs that reflect true control performance.

C. Evaluation of Objective 3: Comparative Analysis

Comparative studies will be conducted against three baseline configurations on identical driving scenarios. First, a GUIDE like mode will implement uncertainty conditioning without evidential outputs. Second, an EPPO like mode will implement evidential critics without localisation uncertainty inputs. Third, standard PPO will serve as a baseline without

either mechanism. All configurations will be evaluated under systematically degraded localisation conditions achieved through controlled sensor noise injection.

Safety will be measured through collision rate and constraint violation frequency [20]. In addition, performance will be assessed via goal completion rate and path efficiency metrics. For methods that produce uncertainty estimates, calibration quality will be evaluated using ECE. The hypothesis is that the proposed unified architecture combining both mechanisms could demonstrate improved safety metrics under degraded localisation compared to ablated baselines. This improvement may arise through maintaining better calibrated uncertainty estimates that enable appropriate risk aware behaviour.

VI. CONCLUSION

In this proposal, evidential policy networks and SLAM uncertainty conditioning are used to provide a unified architecture for autonomous driving. The research tackles the problem of safe navigation under uncertainty at the heart of EDL, RL, and uncertainty aware robots. The main contributions are the use of SLAM covariance matrices as explicit state inputs, which allows the system to actively adapt to localisation reliability, and the application of evidential networks to continuous control policies, which sets this work apart from previous approaches that focused on evidential critics.

The methodology synthesises advancements from related fields, effectively combining the uncertainty conditioned RL of GUIDE/G SAC [11] with the evidential critic methods of EPPO [13]. Furthermore, the suggested architecture seeks to provide calibrated uncertainty estimates straight from the learnt driving policy by combining these ideas under a single framework.

If validated, this approach could support the development of uncertainty aware autonomous systems capable of communicating confidence levels and implementing risk aware decision making. Moreover, the use of the CARLA simulation environment allows for systematic evaluation of these principles, serving as a precursor to potential real world deployment.

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